

NUDT15 polymorphisms alter thiopurine metabolism and hematopoietic toxicity

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Widely used as anticancer and immunosuppressive agents, thiopurines have narrow therapeutic indices owing to frequent toxicities, partly explained by *TPMT* genetic polymorphisms. Recent studies identified germline *NUDT15* variation as another critical determinant of thiopurine intolerance, but the underlying molecular mechanisms and the clinical implications of this pharmacogenetic association remain unknown. In 270 children enrolled in clinical trials for acute lymphoblastic leukemia in Guatemala, Singapore and Japan, we identified four *NUDT15* coding variants (p.Arg139Cys, p.Arg139His, p.Val18Ile and p.Val18_Val19insGlyVal) that resulted in 74.4–100% loss of nucleotide diphosphatase activity. Loss-of-function *NUDT15* diplotypes were consistently associated with thiopurine intolerance across the three cohorts ($P = 0.021$, 2.1×10^{-5} and 0.0054 , respectively; meta-analysis $P = 4.45 \times 10^{-8}$, allelic effect size = -11.5). Mechanistically, *NUDT15* inactivated thiopurine metabolites and decreased thiopurine cytotoxicity *in vitro*, and patients with defective *NUDT15* alleles showed excessive levels of thiopurine active metabolites and toxicity. Taken together, these results indicate that a comprehensive pharmacogenetic model integrating *NUDT15* variants may inform personalized thiopurine therapy.

Thiopurines (mercaptopurine, thioguanine and azathioprine) are widely used anticancer and immunosuppressive agents^{1–8}. In acute lymphoblastic leukemia (ALL), prolonged daily exposure to mercaptopurine is a major component of contemporary treatment regimens and is indispensable for the cure of this disseminated malignancy^{7,9–13}. However, mercaptopurine can cause severe myelosuppression, resulting in frequent treatment disruptions, necessitating extensive supportive care and increasing the risk of life-threatening infections^{6,13–22}. Thiopurines are also commonly prescribed in patients with inflammatory bowel diseases (IBD; for example, Crohn's disease and ulcerative colitis), especially for their steroid-sparing potential and efficacy in the maintenance of remission^{4,6,14–17}. Thiopurine treatment for IBD is associated with substantial hematopoietic toxicity, leading to

discontinuation of therapy in up to 40% of patients and subsequent disease recurrence^{6,18–23}. Therefore, the narrow therapeutic indices of thiopurines point to a strong need for applying evidence-based precision medicine approaches to the use of this class of medications.

As prodrugs, thiopurines are enzymatically converted into thioguanosine triphosphate (TGTP) through multiple sequential anabolic reactions. TGTP is further reduced to deoxythioguanosine triphosphate (TdGTP), which is incorporated into double-stranded DNA (DNA-TG) to trigger futile mismatch repair and, eventually, apoptosis^{1,24–28}. The integration of thioguanine into DNA is widely accepted as one of the principal means of action for thiopurine drugs and a major process responsible for their cytotoxic effects^{8,27–31}. There are multiple pathways that negatively influence thiopurine

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effects: mercaptopurine can be directly anabolized by thiopurine methyltransferase (TPMT) to the inactive methylmercaptopurine, and thiopurine active metabolites can also be catabolized via dephosphorylation of thioguanine nucleotides (TGN)^{32,33}. Therefore, thiopurine cytotoxicity is determined by competition between the activation and inactivation pathways, the components of which are affected by genetic variations. In particular, SNPs in the *TPMT* gene have been shown to cause loss of TPMT enzymatic activity and excessive levels of TGN and to predispose patients to thiopurine-related hematopoietic toxicity across multiple diseases^{23,34–38}. In fact, *TPMT*-guided preemptive thiopurine dose adjustment has been shown to substantially reduce the risk of adverse effects without compromising therapeutic efficacy^{39–45} and is a prototype of precision medicine approaches to individualize drug therapy based on pharmacogenetics. However, substantial toxicity still occurs in some patients with normal *TPMT* activity, suggesting that additional variables, including other genetic variants, may contribute to the between-patient variability in thiopurine metabolism.

Recent genome-wide association studies described a missense variant in the *NUDT15* gene (rs116855232, referred to as c.415C>T or p.Arg139Cys hereafter) that is strongly associated with thiopurine-related myelosuppression in patients with IBD⁴⁶ and in children with ALL⁴⁷. Individuals homozygous for the risk allele encoding p.Arg139Cys were exquisitely sensitive to mercaptopurine and tolerated only 8% of the standard dose, and this *NUDT15* variant alone explained 22% of variance in mercaptopurine tolerance⁴⁷. Constituting a purported purine-specific nucleotide diphosphatase, *NUDT15* is hypothesized to dephosphorylate the thiopurine active metabolites TGTP and TdGTP, thus preventing their incorporation into DNA and negatively affecting the desired cytotoxic effects of thiopurines. However, the exact mechanism by which *NUDT15* regulates thiopurine metabolism has not been experimentally examined, and the functional consequences of *NUDT15* variants are unknown.

In this study, we systematically identified *NUDT15* variants associated with thiopurine disposition and host toxicity, characterized the enzymatic properties of the corresponding proteins and comprehensively investigated the molecular pathways linking *NUDT15* to thiopurine toxicity. Our results suggest that a comprehensive pharmacogenetic model integrating *NUDT15* genetic variants may inform strategies to further individualize thiopurine therapy.

RESULTS

NUDT15 variant discovery and function characterization

To systematically characterize *NUDT15* polymorphisms associated with thiopurine toxicity, we sequenced all exonic regions of this gene in three cohorts comprising 270 children treated on frontline ALL clinical trials in Guatemala, Singapore and Japan (Supplementary Table 1). In total, we identified four coding variants located in exons 1 and 3,

all of which resulted in changes in the amino acid sequence of the *NUDT15* protein (Fig. 1a). The c.415C>T and c.416G>A variants were transition substitutions within the codon for residue Arg139, causing an arginine-to-cysteine and arginine-to-histidine change (p.Arg139Cys and p.Arg139His), respectively. The other two variants affected the Val18 residue: c.52G>A resulted in a valine-to-isoleucine substitution (p.Val18Ile), and c.36_37insGGAGTC led to an in-frame addition of a glycine and a valine residue (p.Val18_Val19insGlyVal). We inferred five *NUDT15* haplotypes with distinct combinations of genotypes at these four variants (referred to as haplotypes *1–*5 hereafter; Fig. 1a). In particular, the p.Val18_Val19insGlyVal allele was in high linkage disequilibrium (LD) with the p.Arg139Cys allele, and the presence of both variants defined the common haplotype *2. Analyses of phased 1000 Genomes Project data identified similar haplotype patterns, and the prevalence of each haplotype varied substantially by ancestry (Supplementary Fig. 1).

To define the functional consequences of the *NUDT15* polymorphisms, we characterized nucleotide diphosphatase activity for each of the four variant proteins (Fig. 1b). Wild-type *NUDT15* efficiently converted the thiopurine active metabolite TGTP to the monophosphate thioguanosine nucleotide TGMP, with an apparent catalytic efficiency ($V_{\max}/K_m$, ratio of maximal velocity to Michaelis constant) of 107.9 ± 0.2 . In contrast, the four variant *NUDT15* proteins showed 74.4 to 100% loss of enzymatic activity, with $V_{\max}/K_m$ of 23.6 ± 0.9 , 27.7 ± 1.1 and 14.9 ± 1.1 for the Arg139His, Val18Ile and Val18_Val19insGlyVal variants, respectively; enzymatic activity was not detectable for the Arg139Cys or Val18_Val19insGlyVal/Arg139Cys variant protein. We observed similar differences in *NUDT15* activity when TdGTP was used as the substrate (Supplementary Fig. 2). Although p.Arg139Cys and p.Val18_Val19insGlyVal individually caused a reduction in *NUDT15* activity, the protein harboring both variants did not show further loss of function as compared to the protein with p.Arg139Cys alone (Fig. 1b), suggesting that haplotype *2 is likely to have similar functional consequences as haplotypes with a single variant (for example, *3). A thermostability assay showed substantially lower T_m (protein unfolding temperature) values for all variant *NUDT15* proteins (42.6 ± 0.1 °C, 43.6 ± 0.2 °C, 52.5 ± 0.3 °C, 41.5 ± 0.1 °C and 54.0 ± 0.1 °C for Arg139Cys, Arg139His, Val18Ile, Val18_Val19insGlyVal/Arg139Cys and Val18_Val19insGlyVal, respectively) as compared to the wild-type protein (56.3 ± 0.3 °C), consistent with perturbation of *NUDT15* conformation by the substitutions (Supplementary Fig. 3).

Figure 1 *NUDT15* genetic variants and their effects on nucleotide diphosphatase activity. (a) Four coding variants in *NUDT15* were identified representing five haplotypes (*1 to *5). (b) Each variant *NUDT15* protein was expressed in *Escherichia coli*, and purified protein was subjected to diphosphatase activity measurement with TGTP as the substrate. WT, wild type; PP_i, inorganic pyrophosphate. (c) Variant or wild-type proteins were combined to determine the level of *NUDT15* activity in patients with different diplotypes. There were no significant differences in nucleotide diphosphatase activity between diplotypes within the intermediate-activity group (green background; $P = 0.73$) or within the low-activity group (red background; $P = 0.19$), as determined using a Kruskal-Wallis test. In b and c, center values (dots) represent the means of triplicate experiments; error bars, s.d.

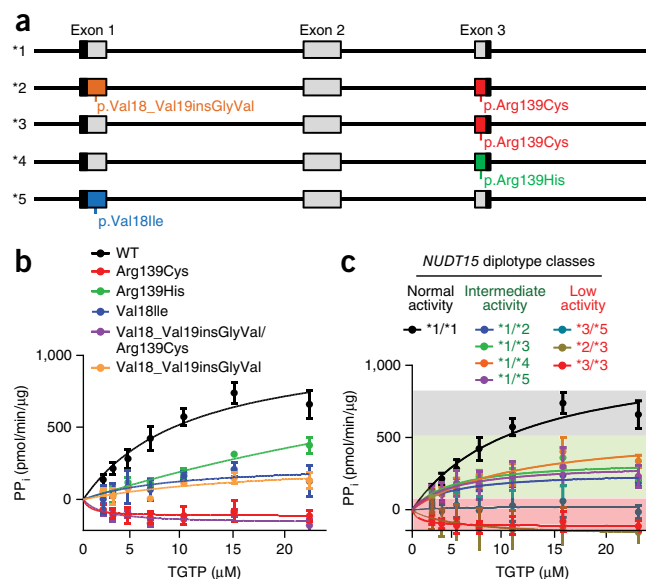
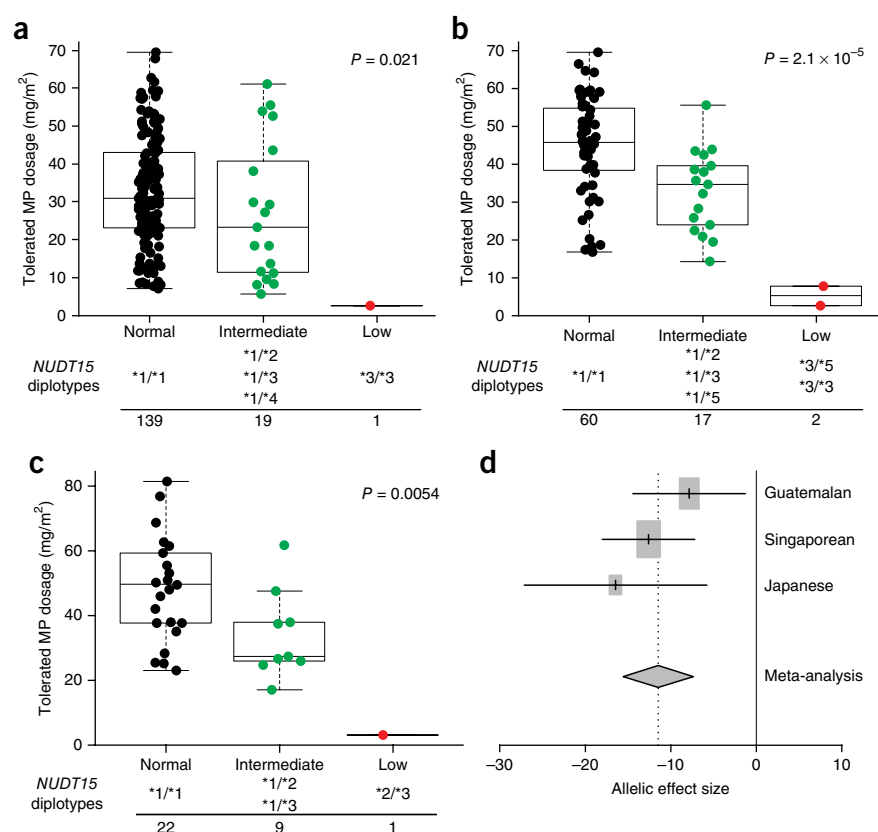


Figure 2 Association of *NUDT15* diplotype with mercaptopurine tolerance during ALL therapy in Guatemala, Singapore and Japan. (a–c) Patients were classified as having normal, intermediate or low *NUDT15* activity on the basis of their diplotype at four coding variants for the Guatemalan (a), Singaporean (b) and Japanese (c) cohorts. Mercaptopurine (MP) dosage was adjusted during maintenance therapy to avoid excessive host toxicities; the tolerated mercaptopurine dosage was defined as the average of at least 14 daily dosages after at least 9 weeks of maintenance therapy. Cases with *TPMT* variants (rs1800462, rs1800460 and rs1142345) were excluded from the analysis. *P* value was calculated using a linear regression test, after adjusting for covariates when applicable. In a–c, each box includes data between the 25th and 75th percentiles, with the horizontal line indicating the median. Similar association analyses were performed to compare tolerated mercaptopurine dosage between the normal- and intermediate-activity *NUDT15* groups ($P = 0.04$, 0.00049 and 0.0033 for a–c, respectively). (d) Meta-analysis combining test statistics from the three cohorts indicated consistent association (Forest plot, $P = 4.45 \times 10^{-8}$), with no significant heterogeneity across the cohorts ($P = 0.34$). Allelic effect size indicates the change in mercaptopurine dosage for every copy of the *NUDT15* risk allele. The length of each horizontal line represents the range of the 95% confidence interval of allelic effect size, with the tick indicating the median (the lengths of the gray boxes are proportional to the weights of each cohort used in meta-analysis). The dotted vertical line denotes allelic effect size in the meta-analysis, with the lateral tips of the diamond representing the 95% confidence interval.



Further, we tested different combinations of variant *NUDT15* proteins to determine enzymatic activity for each of the eight diploypes observed in our clinical cohorts (Fig. 1c). Equimolar mixtures of variant and wild-type *NUDT15* proteins (the equivalent of patients heterozygous at a single variant, for example, *1/*3) showed intermediate activity, indicating a gene dosage effect. In comparison, the mixture of two different variant *NUDT15* proteins (for example, *3/*5) resulted in low activity that was comparable to that for a single variant with the same total protein concentration (for example, *3/*3), suggesting that individuals with compound-heterozygous genotypes are likely to have similar mercaptopurine intolerance as those homozygous for a single *NUDT15* variant. Therefore, we classified patients into three diploypic groups (Fig. 1c): normal activity (*1/*1), intermediate activity (*1/*2, *1/*3, *1/*4 and *1/*5) and low activity (*2/*3, *3/*3 and *3/*5). There were no significant differences in enzymatic activity between diploypes within the intermediate-activity group ($P = 0.73$) or within the low-activity group ($P = 0.19$).

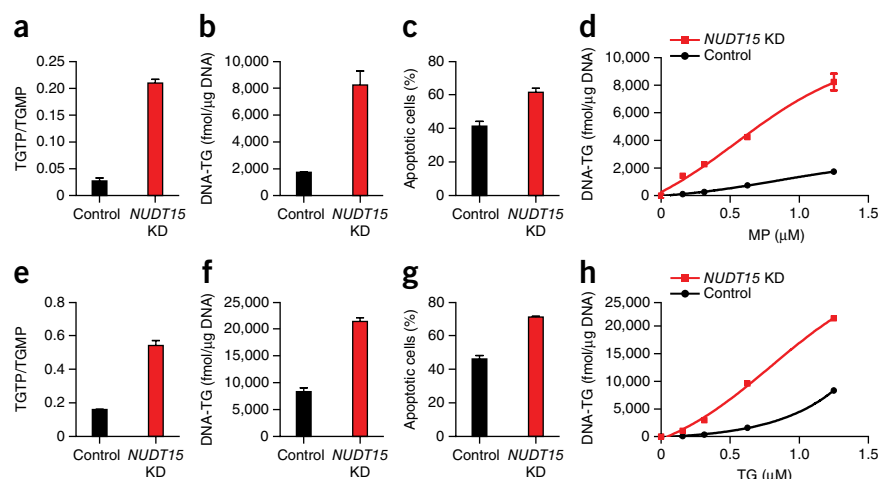
Consistently in the Guatemalan, Singaporean and Japanese cohorts ($n = 159$, 79 and 32 , respectively), mercaptopurine tolerance was highest in patients with the normal-activity *NUDT15* diplotype, followed by those with intermediate activity, and was lowest in patients with the diploypes indicative of low *NUDT15* activity ($P = 0.021$, 2.1×10^{-5} and 0.0054 , respectively; Fig. 2). Of particular note, patients with the *1/*2 diplotype (heterozygous at both p.Arg139Cys and p.Val18_Val19insGlyVal) showed a similar degree of mercaptopurine intolerance as the *1/*3 group (heterozygous at the p.Arg139Cys variant alone; Supplementary Fig. 4), corroborating our observation that single-variant (Arg139Cys)

and double-variant (Val18_Val19insGlyVal/Arg139Cys) *NUDT15* proteins exhibit similar enzymatic activity (Fig. 1b). Additionally, two patients with compound-heterozygous *NUDT15* genotypes (*3/*5 and *2/*3) were exquisitely sensitive to mercaptopurine, with sensitivities on the same scale as for patients homozygous for the variant allele (*3/*3), indicating cumulative effects for the *NUDT15* variants (Supplementary Fig. 4). A meta-analysis combining the three cohorts indicated strong and consistent effects of *NUDT15* variants on tolerated mercaptopurine dosage ($P = 4.45 \times 10^{-8}$, allelic effect size of -11.5 (95% confidence interval, -15.6 to -7.4); Fig. 2d), and we did not observe any statistically significant heterogeneity across the cohorts ($P = 0.34$). Including patients with non-functional variants in *TPMT* (rs1800462, rs1800460 and rs1142345) had minimal effects on the association of *NUDT15* diplotype with tolerated mercaptopurine dosage (meta-analysis $P = 9.96 \times 10^{-10}$ and 9.92×10^{-8} for before and after adjusting for *TPMT* variants, respectively; Supplementary Fig. 5), indicating independent contributions of these two genes to mercaptopurine metabolism and toxicity.

NUDT15 regulates thiopurine activation and cytotoxicity

Because *NUDT15* converts TGTP to TGMP (and TdGTP to TdGMP), we hypothesized that *NUDT15* prevents the incorporation of these thiopurine metabolites into DNA (DNA-TG) and negatively regulates thiopurine activation and, consequently, cytotoxicity. To recapitulate the loss of *NUDT15* activity resulting from genetic variation, we first established stable *NUDT15* knockdown in a human lymphoid cell line (Supplementary Fig. 6). In control cells transduced with scrambled short hairpin RNA (shRNA), mercaptopurine was extensively

Figure 3 Effects of *NUDT15* on thiopurine metabolism and cytotoxicity. (a–h) Cells with *NUDT15* knockdown (*NUDT15* KD; red) were established by lentiviral transduction with *NUDT15*-specific shRNA, and control cells (black) were transduced with non-targeted vectors. The levels of thiopurine metabolites (cytosolic TGMP and TGTP and DNA-incorporated thioguanine (DNA-TG)) were analyzed after treatment with 1.25 μ M mercaptopurine (a,b) or thioguanine (e,f) for 48 h. DNA-TG levels were measured in cells exposed to increasing concentrations of mercaptopurine (d) and thioguanine (h). Cytotoxicity was determined by MTT assay following 72 h of incubation with 1.25 μ M mercaptopurine (c) or thioguanine (g). Mean values are plotted in each panel; error bars, s.d. from triplicate experiments.



metabolized and the ratio of TGTP to TGMP was $0.028 (\pm 0.0052)$, with TGTP representing only 2.7% ($\pm 0.5\%$) of total intracellular TGN. When *NUDT15* expression was downregulated, the level of TGTP increased significantly, with a 7.5-times higher TGTP/TGMP ratio (0.21 ± 0.0067) and a 6.2-times higher percentage of TGTP in TGN ($16.8 \pm 0.3\%$) as compared to control cells, indicating a shift in equilibrium toward TGTP (Fig. 3a). Consequently, DNA-TG levels were markedly higher in *NUDT15*-knockdown cells than in control cells ($8,219.9 \pm 853.6$ fmol/ μ g DNA and $1,734.7 \pm 25.7$ fmol/ μ g DNA, respectively; Fig. 3b), leading to an increase in mercaptopurine-induced apoptosis in these cells (Fig. 3c and Supplementary Fig. 7a). With increasing concentrations of mercaptopurine, there was concomitant elevation of intracellular DNA-TG levels, and the rate at which mercaptopurine was incorporated into DNA was greater in the *NUDT15*-knockdown cells than in control cells (Fig. 3d). Similarly, when we repeated these experiments with thioguanine or azathioprine, the *NUDT15*-knockdown cells consistently showed increased levels of active metabolites (for example, TGTP and DNA-TG) and higher susceptibility to thioguanine- or azathioprine-induced apoptosis (Fig. 3e–h and Supplementary Figs. 7b and 8). Together, these results indicate that *NUDT15* inactivates thiopurine metabolites and is directly related to their cytotoxic effects,

thus providing a clear biological mechanism explaining the susceptibility to thiopurine host toxicity in patients inheriting loss-of-activity variants in *NUDT15*.

NUDT15 variants and thiopurine metabolism *in vivo*

Building upon the *in vitro* results, we next sought to determine how inherited *NUDT15* deficiency influences thiopurine activation in patients by directly monitoring DNA-TG levels in white blood cells from children with ALL receiving daily mercaptopurine treatment. In 32 patients in the Singaporean cohort, DNA-TG levels strongly correlated with the actual mercaptopurine dosage delivered ($P = 0.0024$; Fig. 4a). The positive correlation between mercaptopurine dosage and DNA-TG levels was also validated in the Japanese cohort ($n = 32$, $P = 2.2 \times 10^{-4}$; Fig. 4b). The ratio of DNA-TG levels to mercaptopurine dosage (the amount of DNA-TG converted from every unit mercaptopurine dose) varied significantly by *NUDT15* genotype. In the Singaporean cohort (Fig. 4c and Supplementary Fig. 9a), the average DNA-TG to mercaptopurine dosage ratio was 6.0 ± 2.7 fmol/ μ g DNA/mg mercaptopurine in children with wild-type *NUDT15* (*1/*1). In contrast, children with intermediate-activity *NUDT15* diplotypes (*1/*3 and *1/*5) achieved $8.8 (\pm 5.3)$ fmol/ μ g DNA of DNA-TG for every milligram of mercaptopurine, and this ratio was highest in patients with homozygous or compound-heterozygous *NUDT15* diplotypes (19.6 ± 6.3 fmol/ μ g DNA/mg

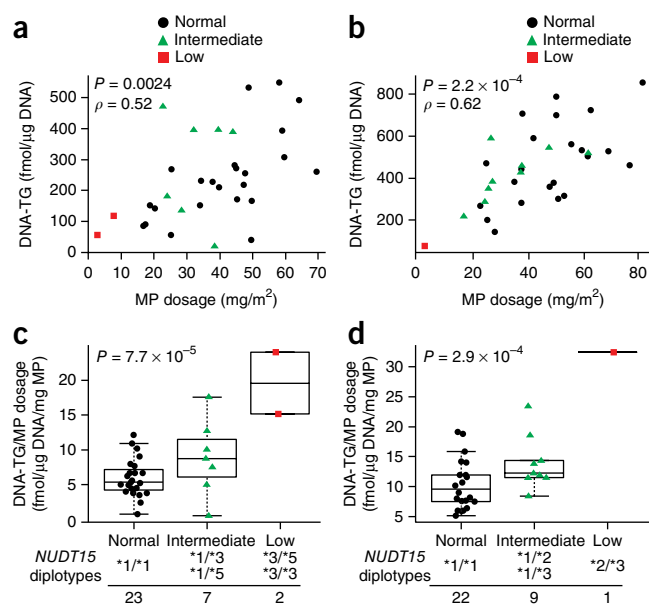


Figure 4 *NUDT15* variants and mercaptopurine metabolism in children during ALL therapy. (a–d) DNA-TG levels were analyzed in Singaporean (a,c) and Japanese (b,d) cohorts. Levels were successfully measured for 64 and 44 samples from 32 cases with wild-type *TPMT* in the Singaporean and Japanese cohorts, respectively. An average DNA-TG level was estimated for each patient. The associations between mercaptopurine dosage and DNA-TG level were evaluated by two-sided Spearman rank test (a,b). DNA-TG level was normalized on the basis of actual mercaptopurine dosage (the average of 14 d) before metabolite measurements and then correlated with *NUDT15* diplotype (as normal, intermediate or low *NUDT15* activity) using a linear regression model (two-sided) (c,d). Similar association analyses were also performed to evaluate the difference in normalized DNA-TG levels between the normal- and intermediate-activity *NUDT15* groups ($P = 0.14$ and 0.039 for c and d, respectively). Meta-analysis combining test statistics from the two cohorts indicated consistent association of *NUDT15* diplotype with normalized DNA-TG level ($P = 2.2 \times 10^{-8}$), without significant heterogeneity between the cohorts ($P = 0.63$). Each box includes data between the 25th and 75th percentiles, with the horizontal line indicating the median and whiskers indicating the maximal and minimal observations that are within 1.5 times the length of the box.

mercaptopurine; $P = 7.7 \times 10^{-5}$). Similarly, in the Japanese cohort (Fig. 4d and Supplementary Fig. 9b), the DNA-TG to mercaptopurine dosage ratio was 9.6 ± 4.1 , 12.3 ± 4.5 and 32.4 fmol/ μ g DNA/mg mercaptopurine for children with normal-, intermediate- and low-activity *NUDT15* diplotypes, respectively ($P = 2.9 \times 10^{-4}$). A meta-analysis combining the Singaporean and Japanese cohorts indicated strong and consistent association of *NUDT15* diplotype with normalized DNA-TG levels ($P = 2.2 \times 10^{-8}$), and we did not observe significant heterogeneity between the cohorts ($P = 0.63$).

DISCUSSION

The current study has comprehensively characterized the effects of inherited variants of *NUDT15* on the metabolism of thiopurine and on the clinical tolerance of thiopurine toxicity. These new findings are important for multiple reasons. First, these *NUDT15* variants are highly penetrant and confer exquisite sensitivity to thiopurines (homozygous carriers can tolerate <10% of the standard dosage), with effect sizes comparable to those of the *TPMT* variants that are clinically implemented to guide thiopurine dose reduction^{46,47}. In fact, in a multiethnic cohort of children with ALL, the *NUDT15* variant encoding p.Arg139Cys alone explained 22% of variance in mercaptopurine tolerance⁴⁷. Second, *TPMT* variants are generally rare among Asian populations⁴⁸ and are thus of limited relevance for guiding thiopurine dosing in these ancestral groups. In contrast, *NUDT15* genetic variation is substantially over-represented in Asians and is the predominant genetic cause in this population of thiopurine toxicity^{46,47}. The unequivocal evidence linking the *NUDT15* variant encoding p.Arg139Cys to thiopurine toxicity (particularly in Asians^{46,47,49,50}) strongly indicates the potential clinical relevance of this variant and also raises the question of whether (or what) clinical action is warranted for at-risk patients.

A critical barrier to integrating *NUDT15* genotypes into the thiopurine dosing algorithm has been the paucity of data to establish the pharmacological basis for dose reduction for patients with *NUDT15* risk alleles. Addressing this knowledge gap, our results from a variety of laboratory model systems and in patients collectively indicate that *NUDT15* deficiency directly results in excessive levels of thiopurine active metabolites (TGTP and DNA-TG) and increased host toxicity. Therefore, reducing thiopurine doses for patients who carry the *NUDT15* variants would likely tailor their exposure to levels of TGTP and DNA-TG that are similar to those in patients with wild-type *NUDT15* receiving standard thiopurine doses. This is a highly plausible strategy for using *NUDT15* genotype to individualize thiopurine therapy to mitigate toxicity, as it is the same principle used for the *TPMT*-based dose adjustments already implemented clinically⁵¹. In fact, in 285 children with newly diagnosed ALL, leukemic cells with intermediate-activity *NUDT15* diplotypes were also significantly more sensitive to thioguanine than those with the wild-type diplotype ($P = 0.03$; Supplementary Fig. 10). Therefore, genotype-guided thiopurine dose reductions in patients with ALL inheriting *NUDT15* risk alleles would likely minimize side effects without compromising antileukemic efficacy. Future clinical studies will be needed to precisely define the optimal dosage of thiopurines in patients with different *NUDT15* diplotypes. The strength of the association between *NUDT15* variants and tolerated mercaptopurine dosage also differed slightly among three ALL cohorts (strongest in Singaporeans and weakest in Guatemalans; Fig. 2). Among these three populations, Guatemalans had the lowest tolerated mercaptopurine dosage even with wild-type *NUDT15* and had smaller differences in mercaptopurine tolerance across the *NUDT15* genotype groups (although the differences were still significant). It is possible that other (as yet

unknown) genetic variants associated with thiopurine toxicity exist uniquely in Guatemalans, potentially contributing to the overall sensitivity of the population to this class of drugs. These hypotheses should be examined in future studies.

Our targeted sequencing of *NUDT15* identified four coding variants, all of which influenced *NUDT15* activity and were linked to thiopurine toxicity (Figs. 1 and 2). Intriguingly, two of these variants both affect residue Arg139 that is located in the $\alpha 2$ α helix at the base of the *NUDT15* substrate-binding pocket⁵². It has been hypothesized that the substitution of arginine with cysteine at this position (p.Arg139Cys) might introduce a disulfide bond and, thus, structural perturbation that interferes with TGTP binding⁵². In contrast, the arginine-to-histidine change caused by the p.Arg139His variant might lead to reduced electrophilicity and compromise substrate interaction. The other two *NUDT15* variants affect residue Val18 within the β sheet at the N terminus of the protein; it is unclear how the substitution and insertion at this position structurally hinder *NUDT15* function. It is also possible that coding variants can affect *NUDT15* protein synthesis and/or degradation and thus indirectly influence enzymatic activity, as we have shown for *TPMT* variants⁵³. However, when ectopically expressed in HEK293T cells, the *NUDT15* variants identified herein showed relatively high protein stability over time *in vitro*, with stability comparable to that of wild-type *NUDT15* (Supplementary Fig. 11); it will be important to document a similar lack of differences in patients with the corresponding *NUDT15* genotypes.

It was originally believed that *NUDT15* functions as a sanitizer to remove damaged nucleotides, for example, the oxo-dGTP generated from radical oxygen species⁵⁴. However, this hypothesis has been challenged by more recent data showing a strong preference of *NUDT15* for thiopurine metabolites over oxo-dGTP⁵². *NUDT15* has also been shown to efficiently hydrolyze dGTP, with potential importance in purine nucleotide homeostasis⁵², but its exact physiological functions remain unknown. It is unclear whether loss-of-function *NUDT15* variants confer susceptibility to any diseases, although there is no evidence in the literature to suggest that this is the case. It is plausible that *NUDT15* deficiency has limited effects on endogenous nucleotide metabolism under normal physiological conditions because of compensation by other NUDT proteins with overlapping enzymatic functions^{52,55}. In contrast, the prominent role of *NUDT15* in thiopurine inactivation signifies the potential importance of the NUDT family of enzymes in the metabolism of other nucleotide analog drugs, which are widely used for diverse human diseases (especially in antiviral therapy)⁵⁶. Therefore, the comprehensive identification of pharmacogenetic variants in NUDT genes may have broad clinical relevance.

Our results from comprehensive *in vitro* and *in vivo* studies strongly indicate that *NUDT15*-related thiopurine toxicity follows an additive genetic mode of inheritance, with the severity of the phenotype proportional to the cumulative number of risk alleles in *NUDT15*. Patients homozygous or compound heterozygous for *NUDT15* risk alleles experience especially excessive mercaptopurine toxicity, as compared to those with intermediate- or normal-activity diplotypes (Fig. 2). This is not unexpected because homozygous-variant *NUDT15* protein is enzymatically inactive (Fig. 1c), and complete loss of this important metabolizing enzyme results in disproportionately high levels of active metabolite and thus toxicity, as also seen in patients with inherited *TPMT* deficiency^{12,35}. The interpretation of *NUDT15* genotype becomes more complicated when a patient carries multiple functional variants. For example, if two variants are on the same haplotype, the patient is likely to have intermediate *NUDT15* activity

because the remaining wild-type copy is still functional (for example, $*1/*2$). In contrast, if each of these two variants affects a different copy of *NUDT15* (compound heterozygous; for example, $*3/*5$), enzymatic activity is completely lost in these patients, similarly to in subjects with a homozygous-variant genotype (for example, $*3/*3$). These two scenarios indicate distinct levels of thiopurine tolerance and can be distinguished only through haplotype-based analyses. Using phased sequencing data from the 1000 Genomes Project, we inferred *NUDT15* diplotypes in major ancestral groups worldwide (**Supplementary Fig. 1**) and experimentally determined the enzymatic activity corresponding to each haplotype. *NUDT15* deficiency (from low- or intermediate-activity diplotypes) was most common in East Asians (22.6%) and was also prevalent in South Asians (evenly distributed in India, Pakistan, Bangladesh and Sri Lanka at 13.6%) and Native American populations (for example, 21.2% in Peruvians and 12.5% in Mexicans). In fact, in our Japanese cohort, approximately one in every three individuals carried risk variant(s) in the *NUDT15* gene, a frequency that is even higher than that of *TPMT* deficiency (for example, one in every 12 Europeans and one in every eight Africans)⁵¹. We have also identified rare *NUDT15* variants in the Broad Institute Exome Aggregation Consortium data set derived from whole-exome sequencing of 60,706 individuals (**Supplementary Table 2**), although the function of these variants and their effects on thiopurine metabolism are unknown.

As previously described⁴⁷, cases with variants in both *NUDT15* and *TPMT* were significantly less tolerant of mercaptopurine than those with risk alleles in only one of these two genes. Therefore, integration of *NUDT15* variants in the thiopurine dosing algorithm may have major implications for Asian populations, whereas *TPMT* variants are most informative for thiopurine toxicity in Europeans and Africans. A polygenic dosing algorithm that incorporates both *NUDT15* and *TPMT* variants would potentially provide a robust approach to personalize thiopurine therapy in major ancestral groups represented in diverse populations worldwide.

URLs. Japanese Pediatric Leukemia/Lymphoma Study Group ALL-B12 (UMIN000009339), <http://www.umin.ac.jp/>; R version 3.0, <http://www.r-project.org/>; 1000 Genomes Project, <http://www.1000genomes.org/>; Broad Institute Exome Aggregation Consortium, <http://exac.broadinstitute.org/>.

METHODS

Methods and any associated references are available in the [online version of the paper](#).

Note: Any Supplementary Information and Source Data files are available in the online version of the paper.

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AUTHOR CONTRIBUTIONS

Supervised research: J.J.Y. Conceived and designed the experiments: T.M., R.N., H.H., K.S., A.E.J.Y., W.E.E. and J.J.Y. Performed the experiments: T.M., R.N., V.P.-A., X.Z., T.-N.L., K.H., J.N., K. Kihira, U.H., R.M., L.L., C.R.N., T.I., Z.C., E.K.-H.C., C.J., Y.L. and M.S. Performed statistical analysis: V.P.-A. and W.Y. Analyzed the data: T.M., R.N., V.P.-A., W.Y., J.N., U.H., R.M., L.L., T.I., C.J., Y.L., M.S., H.H., K.S. and J.J.Y. Contributed reagents, materials and analysis tools: F.A.K., K. Kihira, Y.K., M.K., K. Koh, C.R.N., S.K.-Y.K., Z.C., E.K.-H.C., D.B., H.I., C.-H.P., M.V.R., A.M., H.H., K.S. and A.E.J.Y. Wrote the manuscript: T.M., W.E.E. and J.J.Y.

COMPETING FINANCIAL INTERESTS

The authors declare no competing financial interests.

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ONLINE METHODS

Patients and thiopurine therapy. A total of 270 children treated on frontline ALL clinical trials in Guatemala, Singapore and Japan were included in this study: 266 had newly diagnosed ALL and four cases were diagnosed with mixed-phenotype acute leukemia or lymphoma (**Supplementary Table 1**). The Guatemalan cohort comprised 159 patients (self-identified as Native American or Hispanic/Mestizo) from the LLAG-0707 ALL protocol⁵⁷ at the Unidad Nacional de Oncología Pediátrica (Guatemala City, Guatemala). This protocol included daily mercaptopurine treatment during the maintenance phase with a planned mercaptopurine dosage of 50–75 mg/m² per day including provisions for dose adjustments based on the degree of myelosuppression (target white blood cell count of 1.5–3.0 × 10⁹ cells/l) or infections. The Singaporean cohort included 79 children treated on the MaSpore ALL 2003 and MaSpore ALL 2010 protocols⁵⁸ at National University Hospital (Singapore). The planned daily mercaptopurine dosage was 50 mg/m² and 75 mg/m² for the standard/intermediate-risk and high-risk arms, respectively, and was clinically titrated to a target white blood cell count of 2.0–4.0 × 10⁹ cells/l. The Japanese cohort consisted of 32 children treated on the Japanese Pediatric Leukemia/Lymphoma Study Group (JPLSG) ALL-B12 protocol (UMIN000009339) for newly diagnosed ALL at 13 hospitals and oncology centers in Japan. The standard mercaptopurine dosage during the maintenance phase was 50 mg/m² per day and was adjusted to a target white blood cell count of 2.0–3.0 × 10⁹ cells/l. In all three cohorts, the mercaptopurine dosage was considered to be stable after at least 9 weeks of daily mercaptopurine dosing during maintenance therapy (with appropriate dose titration), following which point the average of the daily dosage over at least 14 d was used to define mercaptopurine tolerance (tolerated mercaptopurine dosage).

Patients were selected on the basis of the availability of germline DNA and the availability of mercaptopurine dosing and tolerance history. Germline DNA was extracted from peripheral blood obtained during clinical remission. This study was approved by the respective institutional review boards, and informed consent was obtained from parents, guardians and/or patients, as appropriate.

NUDT15 sequencing and TPMT genotyping. The coding regions (exons 1–3) of the *NUDT15* gene were first amplified by PCR from germline DNA, followed by Sanger sequencing (primer sequences are provided in **Supplementary Table 3**). Sequence alignment and comparison were performed using the CLC Genomics Workbench (CLC Bio, Qiagen) to identify genetic variants. *NUDT15* variant genotype was coded as 0, 1 or 2 to indicate the number of variant alleles, and the sequencing data were analyzed using PHASE⁵⁹ to infer haplotypes. *TPMT* variants (rs1800462, rs1800460 and rs1142345) were manually genotyped as described previously⁶⁰ for all patients, and individuals with variant genotypes ($n = 20$ and 2 in the Guatemalan and Singaporean cohorts, respectively) were excluded from subsequent analyses unless otherwise indicated.

NUDT15 purification, enzymatic activity measurement and thermostability assay. Human *NUDT15* cDNA (**BC133017**) was cloned into the pCold II expression vector (TaKaRa Bio) with a sequence encoding an N-terminal His tag, as previously described⁵⁴. Missense and insertion variants (p.Arg139Cys, p.Arg139His, p.Val18Ile, p.Val18_Val19insGlyVal and p.Val18_Val19insGlyVal/p.Arg139Cys) were introduced by site-directed mutagenesis using the QuikChange II XL kit (Agilent Technologies). Wild-type or variant *NUDT15* protein was ectopically expressed in *E. coli* BL21 with isopropyl β-D-1-thiogalactopyranoside (IPTG) induction (0.5 mM, 24 h at 15 °C) and purified by affinity chromatography using TALON metal affinity resin (Clontech). Purified proteins were concentrated with an Amicon Ultra 0.5-ml centrifugal filter (Sigma-Aldrich) in 10 mM pH 8.0 Tris-HCl buffer containing 0.1 mM EDTA, 5 mM DTT and 10% glycerol. Protein purity was determined by SDS-PAGE and Coomassie blue staining (**Supplementary Fig. 12**), and the molecular weight of each *NUDT15* protein was confirmed by time-of-flight mass spectrometry using the Waters LCT Premier XE MS (**Supplementary Table 4**).

NUDT15 diphosphatase activity was measured by quantifying the release of pyrophosphate using the PiPer Pyrophosphate Assay kit (Life Technologies). Each *NUDT15* protein (100 ng) was incubated with varying concentrations of

TGTP (1.88 to 21.4 μM) or TdGTP (0.99 to 35.7 μM) at 37 °C for 10 min before heat inactivation (99 °C for 20 min). Free pyrophosphate was then immediately quantified by fluorescence spectrometry according to the manufacturer's instructions, and K_m and V_{max} values were determined by Michaelis-Menten kinetics methods in Prism (GraphPad). Each experiment was carried out in triplicate and was repeated at least three times.

To determine protein thermostability, 1 μg of wild-type or variant *NUDT15* protein was incubated with 10× SYPRO Orange (Molecular Probes), and the mixture was heated from 20 °C to 95 °C in increments of 0.2 °C in the QuantStudio 12K Flex RT-PCR system (Applied Biosystems). Changes in fluorescence were monitored with a charge-coupled device camera, with excitation and emission wavelengths at 490 nm and 575 nm, respectively. T_m values, the temperature midpoint for the protein unfolding transition, were calculated on the basis of the Boltzmann model in Prism.

To assess *NUDT15* protein stability *in vitro*, variant or wild-type *NUDT15* cDNA encoding protein with an N-terminal FLAG tag in the pcDNA3.1 backbone (Invitrogen) was transiently expressed in HEK293T cells (American Type Culture Collection) using polyethylenimine reagent (Polysciences). Cycloheximide (50 μg/ml) was added 48 h after transfection, and *NUDT15* protein levels were monitored after 0, 24 and 48 h by immunoblot analysis with β-actin as the loading control (antibody to FLAG, 2368, Cell Signaling Technology; 1:5,000 dilution).

NUDT15 knockdown and thiopurine metabolism *in vitro*. Lentiviral vectors containing shRNAs specific for human *NUDT15* (TRCN0000050311) or scrambled sequence (SHC016) were purchased from Sigma-Aldrich. Viral particles were prepared using the calcium chloride method in HEK293T cells, and stable knockdown was established in the NALM-6 human lymphoid cell line (German Collection of Microorganisms and Cell Cultures, DSMZ) by lentiviral transduction of shRNA and puromycin selection, following procedures described previously⁶¹. Cell lines were authenticated by karyotyping and cytogenetics and tested negative for mycoplasma contamination. The level of *NUDT15* knockdown was determined by real-time quantitative PCR with human tubulin as the internal control (**Supplementary Fig. 6a**; primer sequences are provided in **Supplementary Table 3**). *NUDT15* expression was also determined at the protein level by immunoblot analysis with β-actin as the loading control (antibody to *NUDT15*, ab122521, Abcam, 1:500 dilution; **Supplementary Fig. 6b**).

To determine thiopurine metabolism *in vitro*, 5 × 10⁶ *NUDT15*-knockdown or control cells were treated with varying concentrations of mercaptopurine or thioguanine at 37 °C for 48 h. Cytosolic thiopurine metabolites (for example, TGTP, TGDP and TGMF) were analyzed with liquid chromatography–tandem mass spectrometry (LC-MS/MS) according to a previously reported method⁶². In parallel, aliquots of mercaptopurine-, thioguanine- or azathioprine-treated cells were processed to extract genomic DNA, and DNA-TG level was measured using LC-MS/MS as described previously⁶³. Mercaptopurine, thioguanine and azathioprine cytotoxicity was determined for *NUDT15*-knockdown and control cells using the colorimetric 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT) assay at the end of the 72-h period of drug exposure. *In vitro* experiments were carried out in triplicate and were repeated at least three times.

Thiopurine metabolism in children with ALL. To investigate *in vivo* thiopurine metabolism, DNA incorporation of TdGTP (the level of DNA-TG) was longitudinally monitored during the maintenance therapy phase in a subset of children enrolled on the MaSpore ALL 2010 and JPLSG ALL-B12 protocols. In the Singaporean cohort ($n = 32$), peripheral blood was collected every 3 months during maintenance, with a maximum of 3, 4 and 5 time points for those on the high-, intermediate- and standard-risk arms, respectively. In the Japanese cohort ($n = 32$), peripheral blood collection was scheduled at weeks 13 and 41 of maintenance therapy. For both cohorts, genomic DNA was extracted from leukocytes within 72 h of blood draw, and DNA-TG level was quantified using LC-MS/MS as described previously⁶³; a detailed mercaptopurine dosing history was recorded for 14 d before each sample collection.

To examine the effects of *NUDT15* variants on leukemic cell sensitivity to thiopurine, we also evaluated thiopurine LC₅₀ (the drug concentration that is lethal to 50% of cells) in 285 children with newly diagnosed ALL. Primary ALL cells were exposed to increasing concentrations of thioguanine for 96 h,

after which cell viability was determined using MTT assays to estimate LC_{50} , as described previously⁶⁴.

Statistical analyses. Each patient was assigned a *NUDT15* 'genetic score' of 0, 1 or 2 indicating low, intermediate or normal *NUDT15* activity, on the basis of his/her *NUDT15* genetic diplotype and experimentally determined enzymatic activity for each diplotype (Fig. 1c). Within each ALL cohort, we first evaluated the association of patient clinical features with tolerated mercaptopurine dosage, using the Wilcoxon rank test (sex as a categorical variable), the Kruskal-Wallis test (ancestry and leukemia type as categorical variables) or the Spearman rank test (age as a continuous variable; **Supplementary Table 1**). The correlation between *NUDT15* genetic score and tolerated mercaptopurine dosage (the average value for each patient if there were multiple measurements) was then tested separately in the Guatemalan, Singaporean and Japanese cohorts, using a linear regression model. Age and sex were included as covariates in the regression model for the Singaporean cohort because they were related to mercaptopurine dosage. A similar statistical model was applied to determine the association of *NUDT15* genetic score and the ratio of DNA-TG to mercaptopurine dosage in the Singaporean and Japanese cohorts. Correlation between tolerated mercaptopurine dosage and DNA-TG level was determined using the Spearman rank test. The Kruskal-Wallis test was used to evaluate the differences in nucleotide diphosphatase activity, tolerated mercaptopurine dosage and normalized DNA-TG level among diplotypes within the intermediate-activity group and among those within the low-activity group. Meta-analyses were also performed combining the test statistics of each cohort using the random-effect model based on the inverse-variance method⁶⁵ with ancestry, age and sex as covariates, and

between-cohort heterogeneity was tested using the Q statistic. All statistical tests were two-sided and were chosen as appropriate according to data distribution; the threshold for statistical significance was defined as $P < 0.05$. R (version 3.0) was used for all analyses unless indicated otherwise.

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